

- Today: (1) Review singular value decomposition, one more example (1)
- (2) Briefly discuss Section 6.5 (won't be on exam) on positive-definite matrices.
- (3) Review topics that will be covered on final exam

(1) Review SVD: A is any $m \times n$ matrix (don't need $m=n$)

Then $A^T A$ and $A A^T$ are both symmetric, have non-negative real number eigenvalues.

For $A^T A$: $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > \lambda_{r+1} = 0, \dots, \lambda_n = 0$

Orthonormal basis of eigenvectors: $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r, \vec{v}_{r+1}, \dots, \vec{v}_n$

Basis for row space $C(A^T)$

Basis for $N(A^T A) = N(A)$

Singular values: $\sigma_i = +\sqrt{\lambda_i}$, so $A^T A \vec{v}_i = \sigma_i^2 \vec{v}_i$

Orthonormal basis

of eigenvectors for: $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_r, \vec{u}_{r+1}, \dots, \vec{u}_m$

$A A^T$

Basis for $C(A)$

Basis for $N(A A^T) = N(A^T)$

$$\vec{u}_i = \frac{1}{\sigma_i} A \vec{v}_i$$

$$A A^T \vec{u}_i = \sigma_i^2 \vec{u}_i$$

Then $A = U \Sigma V^T$

$$\begin{bmatrix} \vec{u}_1 & \vec{u}_2 & \dots & \vec{u}_m \end{bmatrix}$$

orthogonal $m \times m$

$$\begin{bmatrix} \sigma_1 & & & 0's \\ & \ddots & & \\ & & \sigma_r & \\ 0's & & & 0 \dots 0 \end{bmatrix}$$

diagonal-like $m \times n$

$$\begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \dots & \vec{v}_n \end{bmatrix}$$

orthogonal $n \times n$

Example $A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$. Then $A^T A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 5 & 15 \\ 15 & 45 \end{bmatrix}$ (2)

Eigenvalues of $A^T A$: $\begin{vmatrix} 5-\lambda & 15 \\ 15 & 45-\lambda \end{vmatrix} = (5-\lambda)(45-\lambda) - \underbrace{(15)(15)}_{225}$

$= \lambda^2 - 50\lambda + 225 - 225 = \lambda(\lambda - 50) = 0 \rightarrow \lambda = 50, 0$

Only one non-zero singular value: $\sigma_1 = \sqrt{50} = 5\sqrt{2}$

Right singular vectors: $\vec{v}_1$ = unit vector basis of $\lambda=50$ eigenspace,

$\begin{bmatrix} 5-50 & 15 \\ 15 & 45-50 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} -45 & 15 \\ 15 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow y = 3x$

$\rightarrow \vec{x} = c \begin{bmatrix} 1 \\ 3 \end{bmatrix} \rightarrow \vec{v}_1 = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$\vec{v}_2$ = unit ~~the~~ vector basis of null space (of A or $A^T A$)

$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow x = -3y \rightarrow \vec{x} = c \begin{bmatrix} -3 \\ 1 \end{bmatrix} \rightarrow \vec{v}_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$

Left singular vectors: $\vec{u}_1 = \frac{1}{\sigma_1} A \vec{v}_1 = \frac{1}{\sqrt{50}} \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$= \frac{1}{5\sqrt{20}} \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \frac{10}{10\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\vec{u}_2$ = unit vector basis of $N(A^T)$: $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$x + 2y = 0 \rightarrow \vec{x} = c \begin{bmatrix} -2 \\ 1 \end{bmatrix} \rightarrow \vec{u}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

So SVD is: $A = \underbrace{\begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}}_U \underbrace{\begin{bmatrix} 5\sqrt{2} & 0 \\ 0 & 0 \end{bmatrix}}_\Sigma \underbrace{\begin{bmatrix} 1/\sqrt{10} & 3/\sqrt{10} \\ -3/\sqrt{10} & 1/\sqrt{10} \end{bmatrix}}_{V^T}$

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Important geometric property of singular values:

σ_i = amount by which A rescales the length of $\vec{v}_i$,

because $\|A\vec{v}_i\| = \|\sigma_i \vec{u}_i\| = \sigma_i \|\vec{u}_i\| = \sigma_i = \sigma_i \|\vec{v}_i\|$
 $\nwarrow$ unit vector $\nearrow$

In fact, the largest singular value σ_1 gives the maximum amount that A stretches the length of a vector:

$\sigma_1 = \text{maximum of } \|A\vec{x}\| / \|\vec{x}\| \text{ (where } \vec{x} \neq \vec{0} \text{)}$

$\nwarrow$ this maximum is called the "operator norm," denoted $\|A\|$.

For our example: $\left\| \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \right\| = \sigma_1 = 5\sqrt{2}$, $\nwarrow$ maximum stretching factor

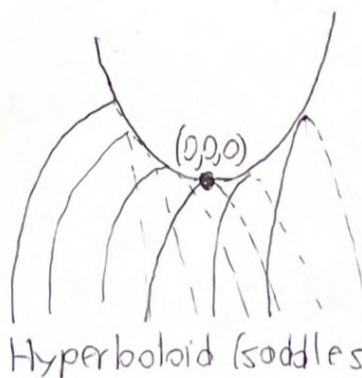
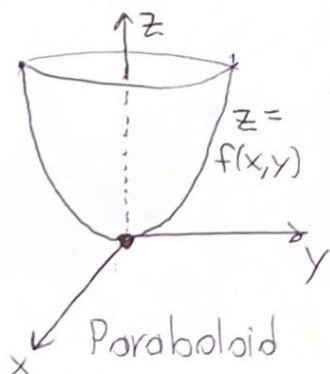
and vectors which get stretched the most are in $\text{span}(\vec{v}_1) = \text{span} \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} \right) = \text{Row space } C(A^T)$.

Section 6.5 Positive-definite matrices

Motivation: Understand quadratic multivariable functions.

$$f(x,y) = ax^2 + 2bxy + cy^2$$

Possible graphs of $f(x,y)$, depending on a, b, c :



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Where does linear algebra come in?

$$f(x, y) = ax^2 + 2bxy + cy^2 = x(ax + by) + y(bx + cy)$$

$$= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} ax + by \\ bx + cy \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix} \underbrace{\begin{bmatrix} a & b \\ b & c \end{bmatrix}}_{\text{Symmetric matrix } S} \begin{bmatrix} x \\ y \end{bmatrix} = \vec{x}^T S \vec{x}$$

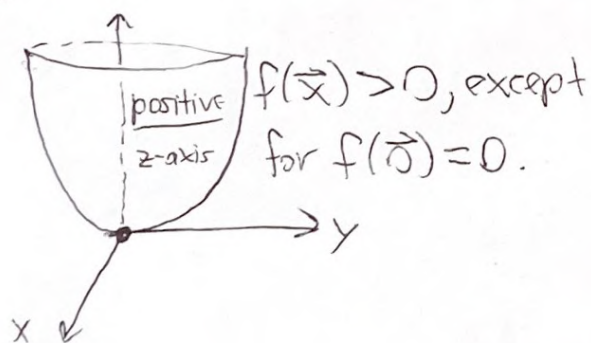
So a two-variable quadratic function has the form

$$f(\vec{x}) = \vec{x}^T S \vec{x} \text{ for some symmetric } 2 \times 2 \text{ matrix } S.$$

Can express n -variable quadratic functions similarly, using symmetric $n \times n$ S .

Definition: A symmetric $n \times n$ matrix S is positive definite if $\vec{x}^T S \vec{x} > 0$ for all non-zero $\vec{x}$ in $\mathbb{R}^n$.

For 2×2 S , this is the same as $f(x, y) = \vec{x}^T S \vec{x}$ having the graph of an upward opening paraboloid:



Is there any practical way to test whether S is positive definite?

There are in fact a few equivalent ways to define positive definite:

Theorem 1: S is positive definite $\iff$ All eigenvalues are > 0 .

Proof If S is positive definite: Suppose λ is an eigenvalue. (5)

$$S\vec{x} = \lambda\vec{x} \rightsquigarrow \underbrace{\vec{x}^T S \vec{x}}_{>0 \text{ (S is positive definite)}} = \lambda \underbrace{\vec{x}^T \vec{x}}_{>0 \text{ (}\vec{x} \neq \vec{0}\text{)}} \rightarrow \lambda = \frac{\vec{x}^T S \vec{x}}{\vec{x}^T \vec{x}} > 0 \checkmark$$

If S has positive eigenvalues: Write $S = Q^T \Lambda Q$

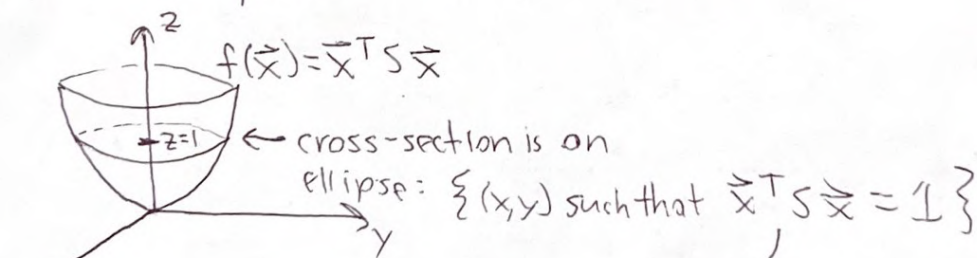
$\nearrow$ positive diagonal $\nwarrow$ orthogonal

Then $\vec{x}^T S \vec{x} = \vec{x}^T Q^T \Lambda Q \vec{x} = (Q\vec{x})^T \Lambda (Q\vec{x})$

$$= \begin{bmatrix} y_1 & y_2 & \dots & y_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \lambda_1 y_1 & \lambda_2 y_2 & \dots & \lambda_n y_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2 > 0 \text{ since } \lambda_1, \lambda_2, \dots, \lambda_n > 0. \checkmark$$

This expression might remind you of the equation of a circle or ellipse.

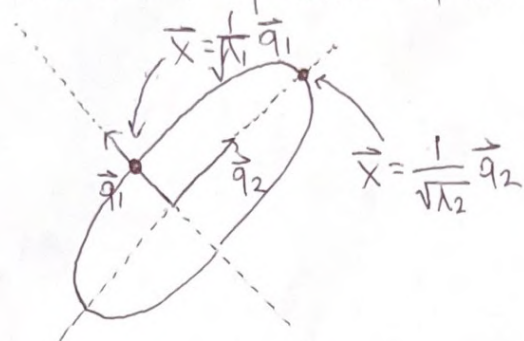


$$1 = \begin{bmatrix} x & y \end{bmatrix} S \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix} Q \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} Q^T \begin{bmatrix} x \\ y \end{bmatrix}$$

$\begin{bmatrix} \vec{q}_1 & \vec{q}_2 \end{bmatrix}$, eigen vector matrix

$$1 = \begin{bmatrix} \vec{x} \cdot \vec{q}_1 & \vec{x} \cdot \vec{q}_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} \vec{q}_1 \cdot \vec{x} \\ \vec{q}_2 \cdot \vec{x} \end{bmatrix} = \lambda_1 (\vec{q}_1 \cdot \vec{x})^2 + \lambda_2 (\vec{q}_2 \cdot \vec{x})^2$$

This is the equation of an ellipse: $\lambda_1(\vec{q}_1 \cdot \vec{x})^2 + \lambda_2(\vec{q}_2 \cdot \vec{x})^2 = 1$ (6)

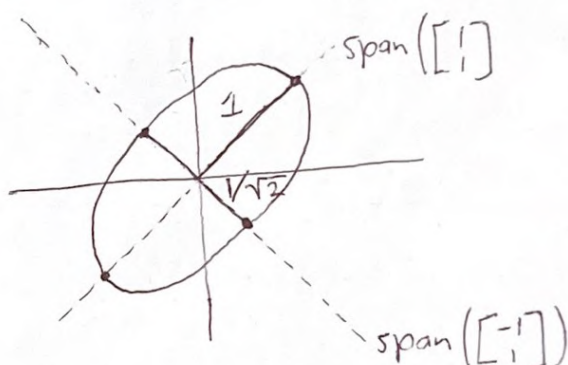


Example $S = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{bmatrix}$

$$1 \left(\frac{x+y}{\sqrt{2}} \right)^2 + 2 \cdot \left(\frac{-x+y}{\sqrt{2}} \right)^2 = 1$$

$$\frac{3}{2}x^2 + 2\left(-\frac{1}{2}\right)xy + \frac{3}{2}y^2 = 1$$

positive eigenvalues are required to make this an ellipse instead of a hyperbola.



Another useful test for positive definite matrices:

S is positive-definite $\iff$ all upper left subdeterminants of S are positive.

2×2 : $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$, need a and $\begin{vmatrix} a & b \\ b & c \end{vmatrix} > 0$.

3×3 : $\begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$, need $a > 0$, $\begin{vmatrix} a & b \\ b & d \end{vmatrix} > 0$, $\begin{vmatrix} a & b & c \\ b & d & e \\ c & e & f \end{vmatrix} > 0$

Same for $n \times n$.

Show this for 2×2 : How to guarantee all $\lambda > 0$?

⑦

$$\begin{vmatrix} a-\lambda & b \\ b & c-\lambda \end{vmatrix} = \lambda^2 - (a+c)\lambda + ac - b^2 = 0$$

$$\rightarrow \lambda = \frac{a+c \pm \sqrt{(a+c)^2 - (ac-b^2)}}{2}$$

For "-" eigenvalue to be > 0 , need $a+c > \sqrt{(a+c)^2 - (ac-b^2)}$

$$\rightarrow (a+c)^2 > (a+c)^2 - (ac-b^2) \rightarrow \boxed{ac-b^2 > 0}$$

det S

If $ac-b^2 > 0$ then $ac > b^2 \geq 0$, so a, c are both > 0 or both < 0

So we need $\boxed{a > 0}$, since otherwise $\frac{a+c - \sqrt{\dots}}{2}$ would be negative.

Example: $S = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ is positive definite because

$$2 > 0, \quad \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3 > 0, \quad \text{and} \quad \begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix} = 2 \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} - (-1) \begin{vmatrix} -1 & -1 \\ 0 & 2 \end{vmatrix} = 2(3) + (-2) = 4 > 0$$

Example what numbers d make $\begin{bmatrix} 1 & 2 & 3 \\ 2 & d & 4 \\ 3 & 4 & 5 \end{bmatrix}$ positive definite?

$$1 > 0 \quad \checkmark \quad \begin{vmatrix} 1 & 2 \\ 2 & d \end{vmatrix} = d-4 > 0 \rightarrow \boxed{d > 4}$$

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & d & 4 \\ 3 & 4 & 5 \end{vmatrix} = \begin{vmatrix} d & 4 \\ 4 & 5 \end{vmatrix} - 2 \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} + 3 \begin{vmatrix} 2 & d \\ 3 & 4 \end{vmatrix} = 5d - 16 - 2(-2) + 3(8-3d) = -4d + 12 > 0 \rightarrow \boxed{d < 3}$$

Inconsistent! No value of d works!

Topics to know for the final:

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Chapter 4: Know what orthogonal vectors/subspaces are.

- Projections: Know how to project vectors onto a subspace and how to compute projection matrices.
- Least squares approximations (application of projection)
- Orthogonal matrices and orthonormal basis. Know how to turn any basis into orthonormal using Gram-Schmidt process.

Chapter 5: Determinants

- Know how to compute using row operations and cofactor expansion.
- Applications to volumes, and cross products.

Chapter 6: Know how to find eigenvalues/eigenvectors. (Hint: on final, eigenvalues will never be more complicated than simple fractions.)

- Know how to diagonalize A (if there is a basis of eigenvectors), and use it to calculate A^N .
- Note = Differential equations will not be on the final exam.

Chapter 7: Know how to compute singular value decomposition of a 2×2 matrix, and know the geometric interpretation of singular values ($\sigma_1 = \max \text{ of } \|A\vec{x}\| / \|\vec{x}\|$)

Key topics to know from earlier in the course:

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Matrix algebra:

- Matrix multiplication
 - Inverses
 - LU decomposition
- } Know how to use these to solve linear equations.

Connections to geometry: Dot products, lengths, angles.

Chapter 3

- Definitions of subspace, null space, column space, row space, left null space.
- Finding all solutions to linear equations.
- Know what it means for vectors to be independent.
- Know how to find bases for subspaces, especially for the four subspaces associated to a matrix.